

Relativistic Future Lookahead & Global Causality Horizon Engine

Mathematical Formalism of Temporal Offsets and Smoothed Time Dilation Transitions

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Abstract

In real-time dynamic digital signal processing (DSP), non-causal operations requiring future signal information $\mathbf{x}(t + \Delta t_{\text{future}})$ present a physical impossibility under standard real-time constraints. This paper presents a novel relativistic solution: the **Global Causality Horizon Offset Engine**. By introducing a dynamic global causality delay $T_{\text{horizon}} = \max_i(\Delta t_{\text{future},i})$ across the entire workspace, the observer present is shifted backward, allowing future-requesting nodes to inspect the live un-delayed stream. Furthermore, to eliminate audio artifacts and transient phase discontinuities during time dilation factor $\gamma(t)$ modulation, we formulate a 1-pole exponential lowpass filter with user-adjustable cutoff frequency f_c and complete bypass control.

1 Introduction

Standard digital audio workstations (DAWs) rely strictly on causal signal processing:

$$y(t) = \mathcal{H}\left(\{x(\tau) \mid \tau \leq t\}\right) \quad (1)$$

In relativistic audio synthesis, nodes may request information from future coordinate times $t + \Delta t$. To realize this without violating physical time progression, the engine establishes a global reference frame offset.

2 Global Causality Horizon Transformation

In our relativistic framework, **any time object generating positive time progression** ($\gamma > 1.0$, such as `[time.warp~]`, `[time.future~]`, or accelerated `[time.transport]`) inherently creates a positive time advance relative to standard 1.0x baseline time:

$$\Delta t_{\text{future},k} = \max\left(0, \frac{\gamma_k - 1.0}{2}\right) \quad (2)$$

Let $\mathcal{N} = \{N_1, N_2, \dots, N_M\}$ be the graph of audio and control nodes. If nodes generate positive time leads $\Delta t_k > 0$, the master causality horizon T_{horizon} is dynamically evaluated as:

$$T_{\text{horizon}} = \max_{N_k \in \mathcal{N}}(\Delta t_k) \quad (3)$$

The coordinate time transformation for standard (causal) nodes N_j is delayed by T_{horizon} :

$$x'_j(t) = x_j(t - T_{\text{horizon}}) \quad (4)$$

Conversely, positive time generating nodes N_k access the un-delayed live stream:

$$x'_k(t) = x_k(t) = x'_k(t + T_{\text{horizon}}) \quad (5)$$

Thus, positive time objects seamlessly project audio and control signals into the future horizon without violating causal execution.

3 Smoothed Time Dilation Lowpass Filter

Rapid changes in the relativistic time dilation factor $\gamma(t)$ can introduce steep pitch jumps and phase discontinuities. To guarantee smooth temporal transitions while granting total user control, the effective time dilation factor $\gamma_{\text{smooth}}(t)$ is governed by a first-order continuous-time differential equation:

$$\frac{d\gamma_{\text{smooth}}(t)}{dt} = 2\pi f_c \left(\gamma_{\text{target}}(t) - \gamma_{\text{smooth}}(t) \right) \quad (6)$$

where $f_c \in [0.1, 50.0]$ Hz represents the user-configurable smoothing cutoff frequency.

3.1 Discrete-Time Recurrence Relation

In discrete sample domain at sample rate $f_s = 1/T_s$:

$$\alpha = 1 - \exp\left(-2\pi \frac{f_c}{f_s}\right) \quad (7)$$

The smoothed time dilation factor is updated per sample n :

$$\gamma_{\text{smooth}}[n] = \begin{cases} \gamma_{\text{target}}[n], & \text{if Bypass is ACTIVE or } f_c = 0 \\ (1 - \alpha)\gamma_{\text{smooth}}[n-1] + \alpha\gamma_{\text{target}}[n], & \text{otherwise} \end{cases} \quad (8)$$

4 Eliminating Glitches During Horizon Expansion via Lowpass Interpolated Delay

A critical challenge occurs when the required causality horizon T_{horizon} increases dynamically (e.g., when a positive time node accelerates from $\gamma = 1.0 \rightarrow 3.0$). An instantaneous step jump in delay time ΔT would cause severe audio buffer tearing, click artifacts, and sample starvation.

To resolve this, our lowpass filter of time f_c is directly applied to the causality horizon expansion rate:

$$\frac{dT_{\text{horizon, smooth}}(t)}{dt} = 2\pi f_c \left(T_{\text{target}}(t) - T_{\text{horizon, smooth}}(t) \right) \quad (9)$$

During the transition phase ($\frac{dT_{\text{horizon, smooth}}}{dt} > 0$), standard nodes read from the global delay buffer using a **4-point Hermite fractional delay interpolator**:

$$x'_{\text{smooth}}(t) = \text{Hermite}\left(x(t - T_{\text{horizon, smooth}}(t))\right) \quad (10)$$

Because the horizon expansion velocity $\frac{dT_{\text{horizon, smooth}}}{dt} \ll 1.0$ is strictly rate-limited by f_c , the delay read pointer moves continuously without buffer tearing. This produces a **smooth, phase-aligned organic pitch bend** during horizon expansion, completely eliminating harsh glitch spikes while achieving the target future horizon.

If the user activates **Bypass Mode** ($f_c = 0$), the smoothing constraint is removed, allowing instantaneous quantum step jumps for experimental glitch sound design.

5 Adaptive Target-Matched Cutoff & Linear Slew Limiting

Rather than using a static lowpass cutoff f_c , the engine dynamically calculates an **adaptive cutoff frequency** f_c^{adaptive} that automatically matches the exact time duration required to travel to the requested future horizon:

$$T_{\text{travel}} = \frac{1}{2} \left| \gamma_{\text{target}} - \gamma_{\text{smooth}} \right| \quad (11)$$

$$f_c^{\text{adaptive}} = \frac{1}{2\pi \cdot T_{\text{travel}}} \quad (12)$$

To complement adaptive lowpass filtering, a **Linear Slew-Rate Limiter** (S_{max}) bounds the maximum acceleration of coordinate time:

$$\left| \frac{d\gamma_{\text{smooth}}}{dt} \right| \leq S_{\text{max}} \quad (13)$$

This dual-stage architecture guarantees:

1. **Constant Doppler Pitch Shift:** Linear slew limiting enforces a constant pitch shift ratio (ratio = $1 - S_{\text{max}}$), delivering an organic, musical pitch glide rather than exponential decay.
2. **Exact Target Arrival:** Adaptive cutoff frequency tuning ensures the system reaches the exact target future horizon T_{future} precisely when the travel phase completes.

6 The Future Preparation Horizon: Latent Pre-Rendering and Instant Execution

The fundamental core principle of our future horizon engine can be stated simply:

Sound from the future may occur; we do not play it yet, but if we need to, we are instantly ready to do so.

Formally, the engine maintains a **Latent Future Preparation Horizon** $\mathbf{x}_{\text{future}}(t + \tau)$ in global RAM memory. While standard audio execution continues at observer present t , future audio calculation threads execute ahead by $\Delta t_{\text{lookahead}}$.

$$\mathbf{x}_{\text{future}}(t) = \mathcal{H}\left(\{x(\tau) \mid \tau \leq t + \Delta t_{\text{lookahead}}\}\right) \quad (14)$$

When a time warp acceleration or future trigger event occurs at t_{event} , no calculation lag or buffer starvation takes place: the future audio stream is **already pre-rendered and phase-aligned**, enabling zero-latency instant execution.

7 Conclusion

This framework provides total mathematical consistency for future lookahead audio prediction and smooth temporal warping, preserving phase coherence while offering instant quantum bypass when raw step transitions are desired.